

Hyun Joon Cha

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Research Interest | *Reinforcement Learning, Data Infrastructure for AI, AI Safety, AI Alignment*

Education

Seoul National University	Mar 2019 – Present
• MS. Mechanical Engineering	
Korea University	Mar 2019 – Feb 2025
• B.S. Chemical and Biological Engineering	

Research Experiences

Renewable Energy Conversion Lab , Seoul National University	<i>Jun 2024 – Present</i>
<i>Graduate Student. Supervisor: Prof. Suk Won Cha</i>	
• Designed Reinforcement Learning-based fuel cell power controller for Fuel Cell Hybrid Electric Vehicles to maximize fuel economy while minimizing fuel cell and battery degradation that led to	
System Bioengineering Lab , Korea University	<i>Nov 2023 – May 2024</i>
<i>Undergraduate researcher. Supervisor: Prof. Min Kyu Oh</i>	
• Engineered a low-cytotoxicity Cas9-BD system for high-GC Streptomyces, reducing off-target effects and enabling multiplexed BGC refactoring that led to a 2.4-fold increase in specialized metabolite production.	

Publications and Conferences

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- Hyun Joon Cha, Donghwan Sung, Hyunjun Yang, Gwangeon Lee, Suk Won Cha*, “Reinforcement Learning-Based Energy Management System for Fuel Cell Hybrid Electric Vehicles Considering Fuel Cell Degradation”, 2025 Korean Society of Automotive Engineers (KSAE) Autumn Conference [1st author]

Selected Honors and Awards

Chemical Engineering Capstone Design Top Award , Korea University	<i>Dec 2024</i>
• Optimized sensor type and deployment strategies in underground garages to detect immediate thermal runaway	

Selected Projects

Global Human Resources Program for Core Technology of Future Mobility , KIAT	<i>May 2026 – Oct 2026</i>
• 6-month scholarship for visiting research	
In-line Defect Battery Detection using EIS , Mona	<i>Jun 2025 – Nov 2025</i>
• Designed a defect battery detection algorithm by analyzing large-scale electrochemical parameter data from battery cell production lines	

Extracurricular Activities and Leadership

TAKSA, Table Tennis Club , Korea University	<i>Mar 2019 – Feb 2025</i>
• President of Table Tennis Club; participated in numerous competitive tournaments	

Skills Summary

Programming Languages: Python, C++, MATLAB

Frameworks/Tools: PyTorch, Tensorflow, Simulink, WSL2, PostgreSQL, DuckDB

Languages: Korean (Native), English (Fluent, iBT TOEFL 5.5, 110)